BAMAT2O7 — PROBABILITY AND STATISTICS

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Assessment-6

Question:

A fruit orchard reported that 15% of the apples harvested last year were rotten. This year, a random sample of 250 apples was collected, and 45 apples were found to be rotten.

At the 0.05 significance level, can we conclude that the proportion of rotten apples in this year's harvest has increased beyond 15%?

Solve the following:

1. State the null hypothesis (H0H_0H0) and alternative hypothesis (H1H_1H1).
2. Identify whether the test is left-tailed, right-tailed, or two-tailed.
3. Calculate the test statistic.
4. Find the critical value at the 0.05 significance level.
5. State your conclusion.

Solution:

Manual way:

```
> View(data)
> p0 <- 0.15
> n <- 250
> x <- 45
>
> p_hat <- x / n
>
> z <- (p_hat - p0) / sqrt(p0 * (1 - p0) / n)
>
> z
[1] 1.328422
> alpha <- 0.05
>
> z_critical <- qnorm(1 - alpha)
>
> z_critical
[1] 1.644854
> if (z > z_critical) {
+     print("Reject H0")
+ } else {
+     print("Do not reject H0")
+ }
[1] "Do not reject H0"
>
```

R's built-in `prop.test()`

```
> prop.test(
+      x = 45,
+      n = 250,
+      p = 0.15,
+      alternative = "greater",
+      correct = FALSE
+ )

        1-sample proportions test without continuity correction

data:  45 out of 250, null probability 0.15
X-squared = 1.7647, df = 1, p-value = 0.09202
alternative hypothesis: true p is greater than 0.15
95 percent confidence interval:
 0.1435263 1.0000000
sample estimates:
   p
0.18
```

Given Data:

- Historical/Hypothesized proportion (p_0): 0.15
- Sample size (n): 250
- Number of successes/rotten apples (x): 45
- Sample proportion ($\hat{p}$):

$$\hat{p} = \frac{x}{n} = \frac{45}{250} = 0.18$$

- Significance level (α): 0.05

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1. Null and Alternative Hypotheses

- **Null Hypothesis (H_0):** The proportion of rotten apples has not increased.

$$H_0 : p = 0.15 \quad (\text{or } p \leq 0.15)$$

- **Alternative Hypothesis (H_1):** The proportion of rotten apples has increased beyond 15%.

$$H_1 : p > 0.15$$

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2. Type of Test

Since the alternative hypothesis contains the greater-than symbol (>), this is a **right-tailed test** (one-tailed).

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3. Test Statistic

Using the one-sample z-test for proportions:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Substitute the given values:

$$z = \frac{0.18 - 0.15}{\sqrt{\frac{0.15(1-0.15)}{250}}}$$

$$z = \frac{0.03}{\sqrt{\frac{0.15 \times 0.85}{250}}}$$

$$z = \frac{0.03}{\sqrt{\frac{0.1275}{250}}} = \frac{0.03}{\sqrt{0.00051}} \approx \frac{0.03}{0.022583}$$

$$z \approx 1.33$$

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4. Critical Value

For a **right-tailed test** at $\alpha = 0.05$:

$$z_{\text{critical}} = z_{0.05} = 1.645$$

(Rejection region: Reject H_0 if $z > 1.645$)

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5. Conclusion

- **Decision Rule:** Compare the calculated test statistic to the critical value:

$$z_{\text{calculated}} = 1.33 < z_{\text{critical}} = 1.645$$

- **Decision: Fail to reject the null hypothesis (H_0)** because the test statistic does not fall in the rejection region (p-value $\approx 0.0918 > 0.05$).
- **Final Statement:**

> *At the 0.05 significance level, there is* **not enough statistical evidence** *to conclude that the proportion of rotten apples in this year's harvest has increased beyond 15%.*

Code for reference-->

```
p0 <- 0.15
n <- 250
x <- 45

p_hat <- x / n
```

```
z <- (p_hat - p0) / sqrt(p0 * (1 - p0) / n)

z

alpha <- 0.05

z_critical <- qnorm(1 - alpha)

z_critical

if (z > z_critical) {
  print("Reject H0")
} else {
  print("Do not reject H0")
}
```